Data pipeline: SEVIRI forms the input, ICNF only selects the labels and scores the output

SATELLITE INPUT

SEVIRI FRP extract

569,019 fire pixels
15-min scans, 2004–2019
FRP [MW], confidence

Clean

896 duplicates removed
sorted, times in UTC
no confidence filter

Native SEVIRI grid

lat/lon → cell (i, j), exact
177 × 106 cells
≈ 3.2 × 4.5 km per cell

Sample index

one sample = one 12 h period
(06–18 / 18–06 UTC)

seen: detection ≤ 5 km,
start − 1 h … + 6 h

target: ≥ 50 ha and seen
2,821 fires
ignore: 10–50 ha, unseen,
± 1 h period boundary
background: everything else

split by year: train 2004–14,
val 2015–16, test 2017–19

Input x

raw detections, 24 h window
6 h history · 12 h · 6 h tail
24 hourly steps × 7 channels

Model

train: 32 × 32 crops
(targets + hard negatives)
predict: full-grid heatmap

loss

points

side branch, not in front of the model: cluster_hotspots.py → events / nuclei
→ nucleus-birth baseline and the persistent-pixel flag

ICNF fire register

813,384 fires, 1980–2023
start, burned area,
ignition point, local time

Clean

duplicates removed
bad ends / positions flagged
local time → UTC

Candidate fires

≥ 10 ha, 2004–2019: 10,911
position → sensor frame
(½ cell NW before Dec 2017)

Target y, weight w

Gaussian σ = 1 cell / target
w = 0 outside Portugal and
≤ 2 cells of an ignore fire

Evaluation

peaks → points (true frame)
matched to ICNF ≤ 5 / 10 km
AP, recall, F1, error

GROUND TRUTH